

Apex BCI: Deterministic Hardware Stack Architecture Specification

1. Executive Summary

The Apex BCI stack is a deterministic, non-invasive Brain-Computer Interface (BCI) architecture designed for mid-range FPGAs (e.g., Lattice ECP5, Xilinx Artix-7). It processes multi-channel EEG signals directly from an ADS1299 analog front-end, extracts motor-imagery (MI) intent in real-time using fixed-point DSP, and translates this intent into stable 2D cursor movements and symbolic keystrokes.

Crucially, the architecture is designed around a **strict hardware safety model** and a **single frame_valid timing barrier**, ensuring zero hidden state, predictable latencies (1-2 frames, ~4-8 ms), and guaranteed system behavior under fault conditions.

2. System Architecture & Data Flow

The pipeline operates sequentially on a single clock domain, triggered by the completion of an SPI frame from the ADC.

Data Flow: ADS1299 MAC $\rightarrow$ IIR Filters $\rightarrow$ Apex v4 Core $\rightarrow$ Kalman Filter $\rightarrow$ Symbolic Decoder $\rightarrow$ Safety/HID Gate

1. **Acquisition:** The MAC reads 216-bit frames from the ADS1299 via SPI.
2. **Filtering & Normalization:** IIR filters isolate specific frequency bands (θ , μ , β).
3. **Feature Extraction:** The Apex core applies adaptive baselines and spatial weights to generate raw 2D intent (U_x , U_y) and a Confidence score (C).
4. **Smoothing:** A steady-state Kalman filter provides bounded noise reduction.
5. **Decoding:** High-confidence vectors are converted into symbolic tokens (SELECT, CLICK, CONFIRM) via dwell FSMs.
6. **Safety & Output:** The hardware safety model gates all outputs. If safety constraints are met, data is shipped via USB HID and MMIO UART for Host Language Model (LM) completion.

3. Core Processing Modules

3.1. ADS1299 SPI MAC (Media Access Control)

- **Role:** Deterministic frame extraction.
- **Mechanism:** Triggers on the falling edge of $DRDY_n$. Shifts 216 bits of data, strips the 24-bit status word, and outputs an array of eight 24-bit signed integers alongside a `frame_valid` pulse.
- **Safety Integration:** Includes a hardware timeout counter. If $DRDY_n$ fails to assert within a 1ms window, a `frame_miss_flag` is raised, triggering an immediate Tier 3 safety halt.

3.2. Apex v4 Neural Core

- **Role:** Extracts raw 2D motor intent and confidence.
- **Math (Fixed-Point Q1.15):**
 - **Adaptive Baselines (EWMA):** Tracks slow signal drift without AC coupling. $B_{\{n\}} = B_{\{n-1\}} + \alpha(F_n - B_{\{n-1\}})$
 - **Spatial Projection:** Applies pre-loaded Left/Right motor weights to the baseline-corrected features.
 - **Intent Generation:** U_x is derived from lateral differences (L - R). U_y utilizes a μ -suppression proxy derived from signal variance.
 - **Confidence (C):** A hardware-efficient sigmoid approximation based on signal-to-noise ratio: $C \approx \text{clamp}((|U_x| - N)/k, 0, 1)$.

3.3. Steady-State Kalman Filter

- **Role:** 2D constant-velocity smoothing.
- **Optimization:** Matrix inversion is prohibitively expensive in fabric logic. Because the measurement noise (R) and process noise (Q) are relatively static post-calibration, the steady-state Kalman gain matrix (K) is precomputed offline and loaded as Q3.13 constants.
- **States:** [u_x , u_y , v_x , v_y , drift].

3.4. Symbolic Decoder

- **Role:** Translates continuous analog intent into discrete, noiseless tokens for the host.
- **Mechanism:** A Finite State Machine (FSM) monitors confidence (C) and position (U_x , U_y).
 - High confidence + sustained threshold crossing $\rightarrow$ increments a **Dwell Timer**.
 - Reaching dwell threshold outputs CONFIRM or SELECT.
 - Short, high-confidence spikes output CLICK.

4. Hardware Safety Model (The "Gate")

To meet potential regulatory requirements (e.g., Health Canada Class II/III), safety is not left to software. It is enforced at the silicon level prior to the HID output.

Tier	State	Trigger Condition	Hardware Action	Recovery
0	Normal	All systems nominal	Passes all data.	N/A
1	Degraded	Low confidence, minor noise	Disables LMS weight updates. Limits output gain.	Auto-recovers when noise drops.
2	Freeze	Variance spike (e.g., jaw clench), ADC saturation	Zeroes out U_x , U_y to freeze cursor. Holds state.	Requires operator explicit clear_latch.

Tier	State	Trigger Condition	Hardware Action	Recovery
3	Hard Stop	Frame timeout, SPI checksum fail, flatline	Disconnects HID output, halts DSP pipeline.	System reset required.

5. Resource and Verification Strategy

5.1. Resource Envelope

The stack is designed to be highly economical, allowing for future expansion (e.g., adding EMG/EOG fusion blocks):

- **Arithmetic:** Features use Q1.15; Accumulators use Q5.11 to Q15.16 to prevent overflow.
- **Utilization (Estimated on Artix-7 100T):** LUTs ~15%, DSPs ~20-30% (heavily dependent on IIR filter stages), BRAM < 5%.

5.2. Verification (Cocotb)

Verification is performed using a Python-based Cocotb testbench. The testing methodology focuses on **Fault Injection**:

1. **Synthetic EEG:** Injecting low-variance sinusoids to prove mathematical convergence.
2. **Artifact Injection:** Injecting 100\times amplitude spikes to verify saturation logic and ensure Tier 2 isolation triggers within 1 clock cycle.
3. **Timeout Injection:** Halting the simulated SPI clock to prove the 1ms watchdog correctly triggers a Tier 3 hard stop.

6. Host Integration (Thought-to-Text)

Because non-invasive scalp EEG has biological bandwidth limits, the system does not attempt raw speech reconstruction. Instead, it relies on a highly reliable **Symbolic + Language Model (LM)** paradigm.

1. The FPGA sends guaranteed, bounded [Token, U_x, U_y, Conf, Tier] packets over USB.
2. The host software renders a 2D virtual keyboard.
3. The user navigates via U_x, U_y and selects letters via the CONFIRM token.
4. A host-side LLM performs predictive completion and error correction, achieving practical communication rates.